

Hydrological Pattern Analysis: Rivers & Reservoirs

A Multi-Sensor Remote Sensing Approach for the Krishna River Basin

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Abstract—This report addresses the critical challenge of quantifying hydrological dynamics in major river systems and reservoir storage units under the dual pressures of climate variability and anthropogenic activity. Focusing on the Krishna River basin and the specific hydrological influence of the Nagarjuna Sagar Dam, the study utilizes a comprehensive multi-sensor remote sensing approach (Sentinel-2 MSI, CHIRPS, GLDAS) to monitor surface water extent, river width, and channel migration from 2018 to 2025, alongside decadal trends extending back to 1995. The methodology introduces a robust algorithmic framework for separating river channels from reservoir water bodies using morphological erosion techniques. Key findings reveal a strong “hydrological bank” effect, where monsoon groundwater recharge serves as a critical predictor for dry-season surface water availability ($r \approx 0.75$). Furthermore, the study uncovers a significant divergence between high-frequency hydrological fluctuations driven by rainfall and low-frequency landscape changes driven by agricultural expansion (+4.93%). This report details the systematic exploration of these patterns, validating the use of remote sensing as a proxy for basin-scale water health and offering a scalable framework for hydrological monitoring in data-scarce regions.

Index Terms—Hydrology, Remote Sensing, Krishna River, Sentinel-2, Groundwater Coupling, Channel Migration, Morphological Erosion.

I. INTRODUCTION: MOTIVATING USE CASES

The ability to accurately monitor, model, and predict the morphological and hydrological behaviors of major river systems is no longer just an academic exercise in physical geography or environmental science; it has profound practical applications across critical sectors of society. This is particularly relevant in developing regions facing rapid climatic and demographic shifts. River systems and reservoirs are dynamic entities that exist in a state of continuous flux, responding to a complex interplay of natural climatic cycles and direct human engineering. However, in many regions, there remains a critical lack of long-term, integrated data that explicitly links these drivers to the physical changes observed on the ground.

A. Urban Planning and Floodplain Management

By forecasting how river channels migrate and where floodplains expand, city planners and local authorities can optimize land-use zoning and infrastructure development. The analysis of channel migration, which this report identifies as averaging over 20 meters per year in active zones, is crucial for preventing infrastructure development in high-risk erosion areas. Understanding the historical bounds of the river allows for the delineation of safety buffers, ensuring that urban

expansion does not encroach upon dynamic river banks that are liable to shift during extreme events [1].

B. Reservoir Operations and Water Security

The operation of major dams, such as the Nagarjuna Sagar, requires precise knowledge of inflow patterns and storage dynamics. This study demonstrates that reservoir surface area acts as a visual proxy for the total water storage of the basin, including invisible groundwater reserves. Understanding this coupling allows reservoir managers to optimize release schedules not just based on current inflows, but on the “carry-over” storage potential indicated by groundwater levels.

C. Agricultural Sustainability and Irrigation Planning

With agriculture identified as the dominant driver of land-use change in the basin, understanding the correlation between irrigation availability (surface water) and agricultural expansion is vital. The report highlights how agricultural encroachment into floodplains and wastelands alters the local hydrological cycle.

II. LITERATURE REVIEW

A. Advancements in Water Body Detection

The foundational work in this field relies heavily on spectral indices. The Normalized Difference Water Index (NDWI) and its modification, the Modified Normalized Difference Water Index (MNDWI), remain the industry standard for delineating open water. As utilized in this study, these indices exploit the strong absorption of near-infrared (NIR) and shortwave-infrared (SWIR) radiation by water, contrasting it with the high reflectance of vegetation and soil. While recent literature highlights their utility, they struggle in detecting sediment-rich water common during the monsoon season. This study builds on these methods by applying them across a multi-year time series to smooth out such anomalies.

B. Land Use and Hydrological Interaction

Integrating Land Use and Land Cover (LULC) dynamics with hydrological changes is a key area of research. Studies such as those by Debnath et al. (2023) regarding LULC dynamics have established the importance of integrating geospatial and statistical methods [1]. This report builds upon such frameworks by applying them to the Krishna basin, specifically isolating the interactions within a 5km buffer of the water bodies.

C. Subsurface-Surface Coupling

A significant gap in traditional remote sensing of rivers is the inability to “see” groundwater. However, recent approaches using land data assimilation systems (like GLDAS) have allowed researchers to use soil moisture as a proxy for groundwater health. This report contributes to this growing body of literature by empirically demonstrating the “Lagged Effect”—a strong statistical link between monsoon groundwater recharge and subsequent dry-season surface water extent.

III. PROBLEM STATEMENT

The core objective of this project is to quantify the dynamics of river channels and reservoir areas, explicitly linking observed changes to climatic trends and anthropogenic pressures. This task is inherently difficult due to several factors:

- **Climate Variability:** River systems are under increasing pressure from erratic rainfall patterns. The distinction between a “normal” monsoon and an extreme rainfall event can cause massive fluctuations in river width.
- **Anthropogenic Interference:** Direct human activities, such as dam construction and land-use change, obscure natural hydrological signals.
- **Data Limitations:** Developing a consistent time series is hindered by the limitations of optical satellite remote sensing, primarily cloud cover during the monsoon season.
- **Complex Morphology:** Distinguishing between a “river” (flowing, narrow) and a “reservoir” (static, wide) is not trivial in a complex basin where backwaters extend kilometers upstream.

IV. DATA AND METHODOLOGY

A. Data Sources

The analysis relies on a multi-sensor approach, integrating high-resolution optical data with climatic and hydrological datasets (Table I).

TABLE I
DATASET SPECIFICATIONS AND UTILITY

Dataset	Specification	Role in Project
Sentinel-2 MSI	10 m Resolution; Optical/Multispectral	Detailed mapping of river width, reservoir area, and LULC near water bodies.
CHIRPS	Rainfall Estimates (2018-present)	Assessing climatic forcing and monsoon strength.
GLDAS	Land Data Assimilation; Soil Moisture	Proxy for groundwater; essential for evaluating dry-season depletion and recharge.
Bhuvan LULC	Land-cover classification	Detecting broad categories of land-use change.

B. Rationale for Satellite Choice

Sentinel-2 was selected over Landsat 8/9 for the primary analysis period (2018-2025). This choice was driven by Sentinel-2’s 5-day revisit cycle, which significantly increases the probability of acquiring cloud-free images during the

monsoon, and its 10m spatial resolution, which is critical for resolving river banks and measuring channel migration within the $100km \times 100km$ Area of Interest (AOI).

C. Algorithm: Feature Separation

A core methodological contribution of this study is the automated separation of river channels from reservoirs using morphological erosion.

1) *Logic and Implementation:* The algorithm transforms raw binary water masks into classified geometries:

- 1) **Noise Removal:** The script ignores small water patches ($< 1km^2$).
- 2) **Morphological Erosion:** A negative buffer operation is applied to the water body geometries.

$$O_{eroded} = O_{original} \ominus K_x \quad (1)$$

Where K_x is a structural element of radius x meters.

3) Differentiation:

- *Rivers:* Being narrow linear features ($width < 2x$), they fragment or vanish.
 - *Reservoirs:* Being wide bodies, they retain a substantial core.
- 4) **Reconstruction:** Objects retaining a core are classified as reservoirs; separated geometries are then used to generate distinct shapefiles for analysis.

D. River Width Derivation

Direct measurement of river width across thousands of points is computationally expensive. We employed an area-based estimation method. Assuming a river segment can be modeled as a long rectangle where Length l is much greater than Width w ($w \ll l$):

$$Area = w \times l \quad (2)$$

$$Perimeter \approx 2 \times (w + l) \approx 2l \quad (3)$$

Substituting $l \approx Perimeter/2$ into the area equation:

$$w = \frac{2 \times Area}{Perimeter} \quad (4)$$

This approximation smoothes out local bank irregularities, providing a robust basin-scale metric.

E. Channel Migration Metrics

Migration rates were calculated by extracting the centerline of the river for 2020 and 2025. The Local Migration Rate is defined as:

$$Rate = \frac{min_dist(Point_{2020}, Boundary_{2025})}{\Delta t} \quad (5)$$

Where $\Delta t = 5$ years.

V. METRIC: GEO-STATISTICAL EVALUATION

The primary metric for evaluating the relationship between hydrological variables is the Pearson Correlation Coefficient (r):

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \quad (6)$$

Values of r near 0.75 observed in this study indicate a strong predictive relationship between groundwater recharge and surface availability.

VI. RESULTS, OBSERVATIONS, INFERENCES

A. Seasonal Hydrological Dynamics

The time-series analysis of water surface area reveals distinct seasonal patterns. As shown in Fig. 1, the reservoir area and river width exhibit variance that correlates with, but is distinct from, the monsoon cycle.

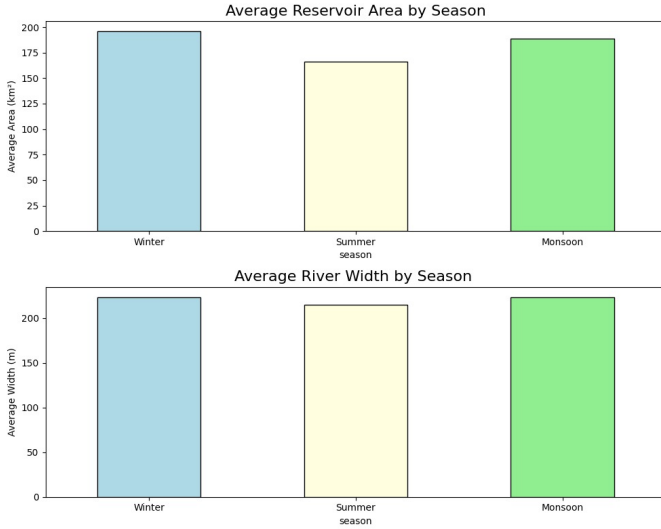


Fig. 1. **Average Reservoir Area and River Width by Season (2018-2025).** Note that Winter reservoir area marginally exceeds Monsoon area, indicating active dam management (flood buffering in Monsoon, retention in Winter).

Table II and Fig. 1 highlight a counter-intuitive finding: Winter reservoir area often exceeds or matches Monsoon levels.

- **Observation:** Winter area is highest (marginally).
- **Inference:** This indicates active reservoir management. The dam is likely not kept at maximum capacity during the monsoon to ensure flood buffer space. Once flood risk subsides in Winter, the dam is closed to retain water.

B. Channel Migration Results

The channel migration analysis (comparing the 2020 boundary to the 2025 boundary) quantified the physical movement of the river banks. A comprehensive visualization of this analysis is presented in Fig. 2.

Fig. 2 illustrates the spatial variability of the river's movement:

TABLE II
SEASONAL SURFACE AREA TRENDS

Season	Area Trend	Observation
Summer	Lowest	High evaporation, scarce rainfall.
Monsoon	High	Recharged by inflow; data skewed by cloud cover.
Winter	Highest	Dam closed to retain water; NE Monsoon input.

- 1) **Centerline Comparison:** The deviation between the 2020 (blue) and 2025 (red) centerlines is evident, particularly in the meandering sections.
- 2) **Deviations:** The histogram confirms a Mean Migration Rate of approx 20.18 m/year (Total $\approx$ 100m over 5 years). The presence of a long tail in the distribution (refer to Hausdorff Distance) indicates localized “hotspots” where erosion is significantly higher (up to 66 m/year).

These metrics suggest that erosion control measures should be targeted specifically at these high-velocity hotspots rather than applied uniformly.

C. The “Hydrological Bank”: Groundwater-Surface Coupling

The study confirms a “lagged effect” where the intensity of the Monsoon groundwater recharge serves as a predictor for the health of the river in the following dry season.

- Monsoon Recharge vs. Dry Season River Area: $r = 0.75$ ($p = 0.052$)

Groundwater acts as a “hydrological bank.” During the dry season, this stored water is slowly released as baseflow, sustaining river width.

D. Rainfall Attribution and Anomalies

Correlating rainfall (CHIRPS) with water metrics revealed limitations in optical remote sensing (Table III).

TABLE III
SEASON-WISE PEARSON CORRELATIONS WITH RAINFALL

Season	Width	Res. Area	Explanation
Monsoon	-0.63	-0.06	<i>Anomaly:</i> Cloud masks and saturation.
Summer	0.18	0.03	Flows dominated by dam releases.
Winter	0.17	0.84	Strong inflow-storage relationship.

The negative correlation in Monsoon (-0.63) is termed the “Monsoon Paradox,” where the wettest days are the least visible due to clouds, leading to potential underestimation of water extent in optical data.

E. Decadal Trends (1995-2025)

Using Landsat 5 and Sentinel-2, a consistent upward trend in reservoir area was observed ($277.89km^2$ in 1995 to $512.28km^2$ in 2025). This is attributed to improved flow regulation, higher carry-over storage strategies, and canal modernization.

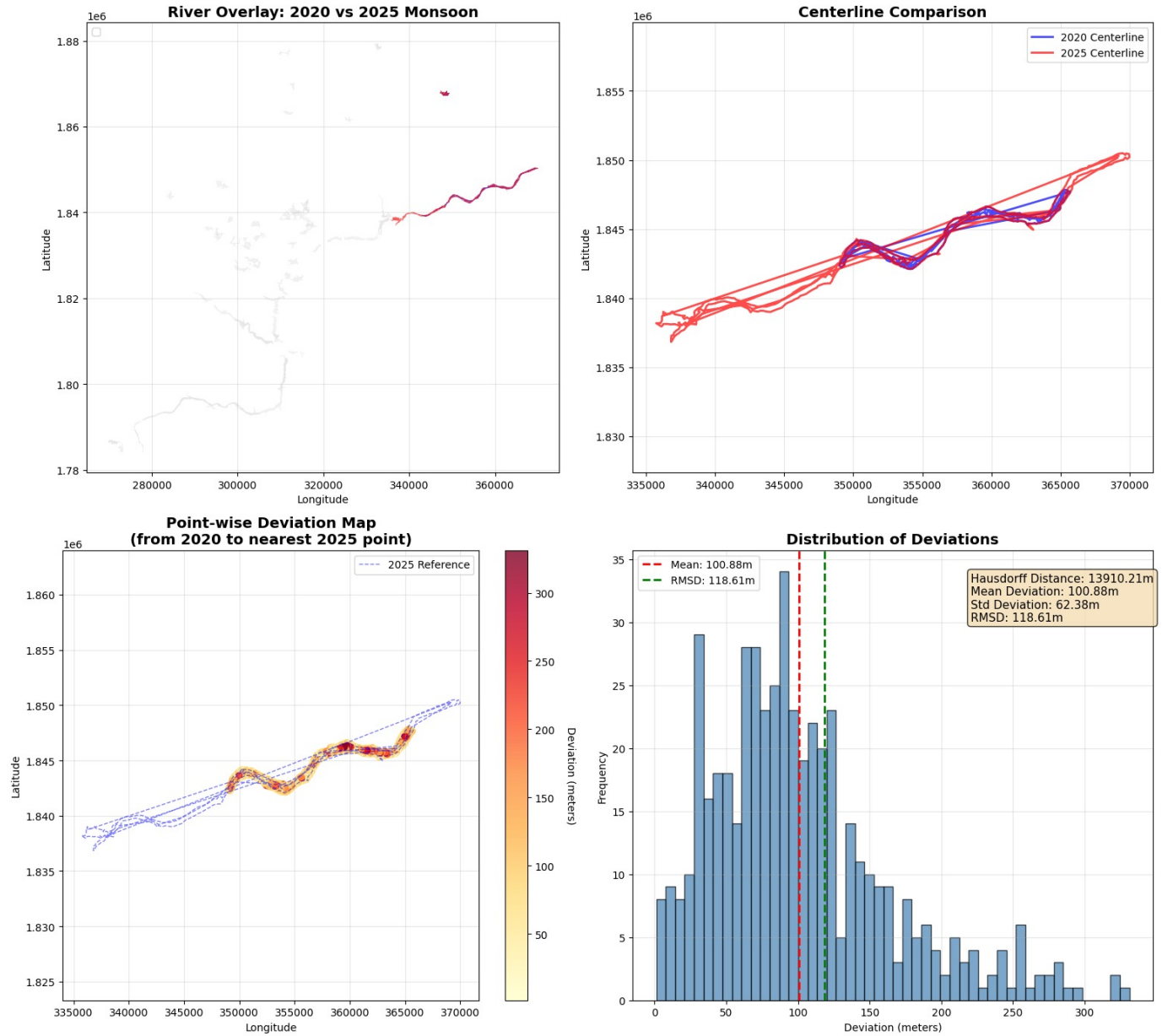


Fig. 2. **River Channel Migration Analysis (2020 vs 2025).** (Top Left) River Overlay showing the shift in boundaries. (Top Right) Centerline comparison indicating lateral shifts. (Bottom Left) Point-wise deviation map highlighting erosion hotspots in red. (Bottom Right) Distribution of deviations, showing a mean migration of 100.88m over the study period.

VII. CHALLENGES AND FUTURE IMPROVEMENTS

A. Challenges Encountered

- **Cloud Cover and Data Gaps:** The “Monsoon Paradox” hinders real-time monitoring during peak flow.
- **Sensor Inconsistency:** The resolution gap between Landsat 5 (30m) and Sentinel-2 (10m) introduces uncertainty in long-term trend analysis.
- **Reservoir Geometry:** Using surface area as a proxy for volume has limitations due to the non-linear relationship between depth and area.

B. Future Improvements

Future work should focus on building predictive models to forecast river migration paths based on climate scenarios and developing flood risk maps by combining historical water extent data with population layers.

VIII. CONCLUSION

This project conducted a comprehensive evaluation of hydrological patterns in the Krishna River basin. The results indicate that the river system acts as a complex, coupled system where groundwater recharge plays a decisive “banking” role in sustaining dry-season flows. The analysis of land-use trends

highlights the relentless expansion of agriculture (+4.93%), which continues largely independent of short-term hydrological stress. The clear decadal trend of increasing reservoir storage points to an increasingly regulated and engineered water system. These findings suggest that projects enhancing groundwater recharge will have a direct, positive impact on keeping rivers and reservoirs fuller during critical dry months.

REFERENCES

- [1] J. Debnath et al., "LULC dynamics in Brahmaputra River basin: Integration of geospatial and statistical methods," 2023.